Vikas Gupta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.79
SDDT Inter college Gorakhpur	Class XII (CBSE)	2014	75.20%
Geeta international public school	Class X (CBSE)	2012	9.80%

SCHOLASTIC ACHIEVEMENTS

- Secured the second highest CGPA in first year M.Tech in Integrated Design and Manufacturing.
- Ranked among the top performers in B.Tech Mechanical Engineering.

SKILLS

CAD & Design: 3D CAD, CAD/CAM, DFMA

Simulation: FEA, FEM, Abaqus/CAE, ANSYS Fluent, CFD, Contact Mechanics

Manufacturing: Additive Manufacturing, Wire EDM, CNC Milling, Laser Processing

Programming & ML: Python, MATLAB, Machine Learning, XGBoost, SHAP, NumPy, Pandas

PROJECTS

Fabrication of Microheaters using Laser-Induced Forward Transfer (LIFT) on Transparent Glass

M.Tech Thesis: IIT Hyderabad | Ongoing

- **Developing** copper nanoparticle-based transparent microheaters using **Laser-Induced Forward Transfer (LIFT)** on glass substrates for transparent heating applications.

Comparative Study of Weld Fumes Measurement in GMAW/FCAW/SMAW Process

B.Tech Mechanical Engineering

- **Designed** a mild-steel fume-testing chamber for comparative emission analysis of **GMAW, FCAW and SMAW**, quantifying **SO, NOx, CO and metal-fume concentrations** in accordance with **IS-11255**.

Finite Element Analysis of Eyelid-Cornea Interaction during Blink

IIT Hyderabad | Abaqus/CAE

- **Developed** a 3D finite-element model of eyelid-cornea interaction using **Abaqus/CAE, contact mechanics and C3D8R elements**, obtaining **24.2 kPa von Mises stress versus 20 kPa reported in literature**.
- **Validated** stress and displacement responses against peer-reviewed literature across multiple blink angles, supporting applications in **contact-lens and dry-eye treatment design**.

CFD Analysis of NACA 0006 Aerofoil Performance

ANSYS Fluent

- Simulated flow over a **NACA 0006 aerofoil** using **ANSYS Fluent** and the **k- SST turbulence model** to characterize **lift, drag, and stall behavior**, running **10 simulation cases** across two Reynolds numbers and five angles of attack, identifying **stall angles of 8° and 16°**.
- Analyzed **lift-to-drag ratio (Cl/Cd)** trends across spanwise planes to pinpoint peak aerodynamic efficiency, comparing **pressure coefficient distributions** at both velocities.

Predictive Maintenance AI (GitHub)

XGBoost, SHAP, Streamlit, Machine Learning

- **Built** an end-to-end **Predictive Maintenance ML system** using **XGBoost, Python, SHAP & Streamlit** on **10,000 records**, achieving **0.984 ROC-AUC** and **0.884 PR-AUC** through feature engineering and class-imbalance handling.
- **Optimized** the decision threshold to **0.75** (**93.2% precision, 80.9% recall, 86.6% F1**) and **deployed** an explainable application with **SHAP-based failure-risk analysis**.

WORK AND TRAINING

- Completed a **6-week summer training** at **Crown Honda** (Honda dealership) under the supervision of the **Training Manager**, gaining exposure to **automotive sales and service operations**.
- **Completed** a 4-week vocational training at **NTPC Dadri**, gaining exposure to **coal-handling systems, material-handling operations and power-generation workflows** in a large-scale industrial environment.

RELEVANT COURSES

- **Design Domain :** Finite Element Method, Computational Fluid Dynamics, FEM Lab, Elasticity and Plasticity
- **Manufacturing Domain :** Additive Manufacturing, DFMA, CAD/CAM, Industry 4.0, Scaling Law and multiple scale manufacturing, Advanced Material Joining Processes
- **AI/ML** : Machine Learning and its Application

POSITIONS OF RESPONSIBILITY

- Assisted as **Teaching Assistant** for **Wire EDM** lab, guiding students on **2D profile cutting** and analyzing **surface roughness** across varying parameters (current, pulse-on time, wire feed) and **CNC Milling** lab, supervising **work locating, centering, and cutting operations** with focus on machine setup and safety.

EXTRACURRICULAR

- Winner in Institute Football League ,Winner in Institute Hockey League